Richard Liao

Mechanical Engineering Student – University of Waterloo

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■ Projects

SolidWorks 4-Cylinder Engine Assembly
September 2026

Personal Project

- Modelled and assembled a 4-piston engine in SolidWorks, applying advanced mechanical mates to constrain engine components and allow for realistic dynamic animation
- Enhanced proficiency in bottom-up assembly management, constructing and combining 6 separate engine components into a complete multi-part assembly

Custom Door Latch & Striker Bolt Mounting Solution
October 2026

Midnight Sun Solar Car Team

- Designed and modelled custom door latch and striker mounts in SolidWorks based on packaging constraints, optimizing geometry for rapid installation and FDM 3D Printing
- Integrated standard fasteners and a flanged heat-set threaded insert in reverse orientation into the assembly, preventing insert pull-out under tensile loads

The Demon Slayer
January 2026

ICS4U Capstone Project

- Designed and delivered a fully playable survival game featuring 6 unique core combat abilities and game mechanics, implemented in Java utilizing object-oriented programming architecture
- Collaborated and communicated with a partner, ensuring organized workflow and meeting strict academic deadlines

■ Technical Skills

Design & CAD: AutoCAD & SolidWorks

Programming: Python, Java

Simulations: Finite Element Analysis

■ Experience

Trailer Music Composer
December 2022 – PRESENT

Really Slow Motion LLC. *Markham, ON*

- Scored 5+ custom projects for major Hollywood campaigns and clients while strictly following technical specifications and time-sensitive briefs
- Collaborated closely with music supervisors, editors, and producers to manage iterative feedback cycles, ensuring efficient project turnovers

Pool Lifeguard
September 2024 – June 2026

City of Markham *Markham, ON*

- Monitored pool facility to mitigate risk and enforce safety regulations, serving as part of a first aid response team
- Provided clear, effective communication to patrons regarding policies, general inquiries, and during high-pressure emergency situations

■ Education

University of Waterloo
Sept 2026 – Apr 2031

Bachelor of Applied Science, Mechanical Engineering (Co-op) *Waterloo, ON*

- Relevant Coursework: Mechanics, Introduction to AutoCAD / SolidWorks, Engineering Drawing Conventions